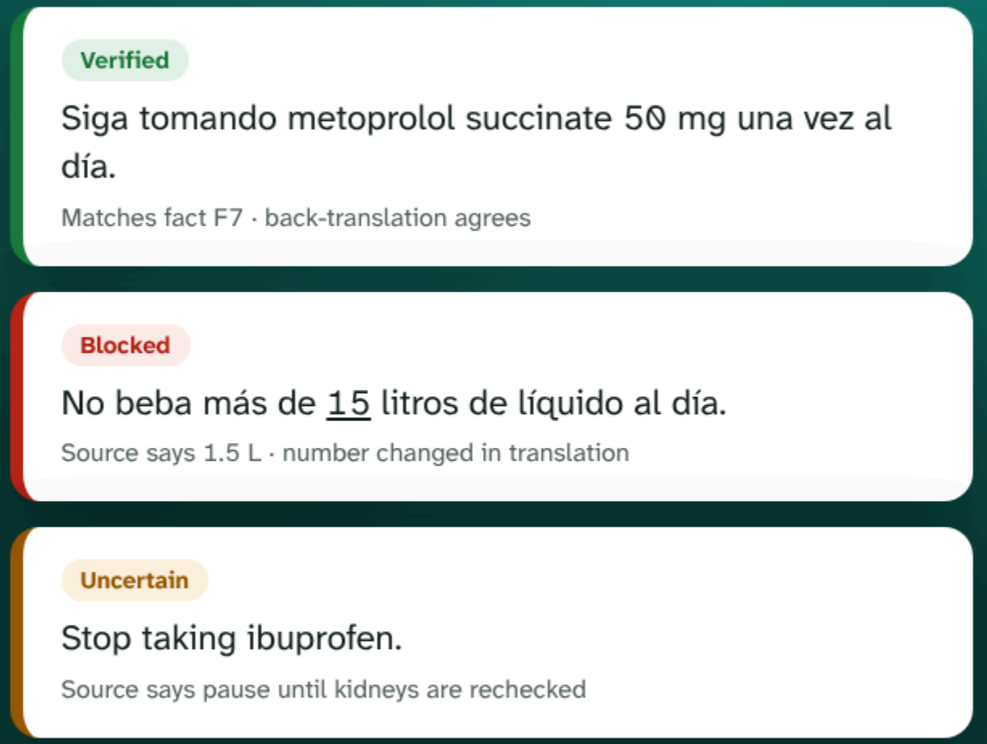
Homeward

Discharge instructions patients can actually follow

Plain language, the patient's own language, every sentence checked against the chart, and a human on every risky line.

Open model on AMD Instinct MI300X · vLLM · ROCm



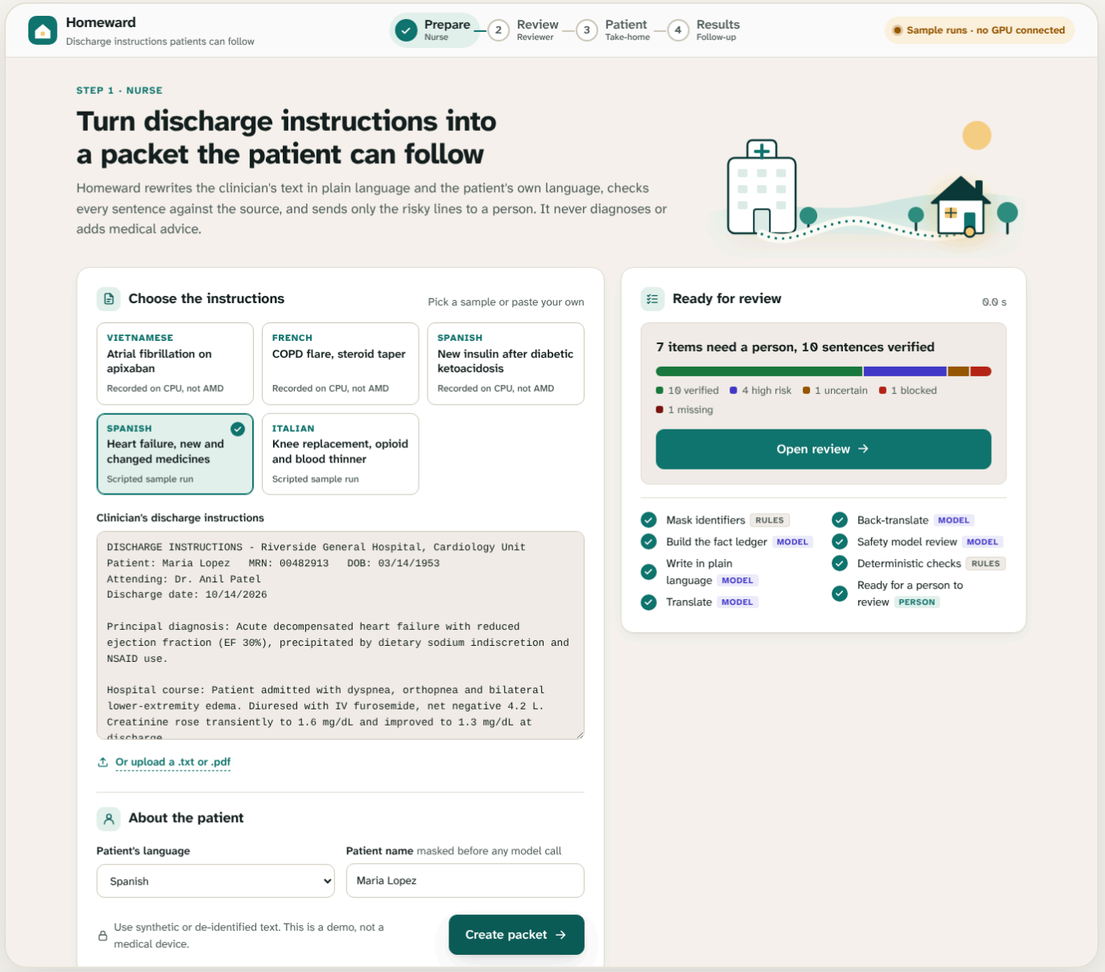
THE PATIENT

Maria, 72, going home after heart failure

Five medicine changes in one day: one new, one doubled, one paused, one stopped, one swapped with a 36-hour gap.

The instructions are in English, written for clinicians. Maria reads Spanish. Her daughter will manage her pills.

If Maria keeps taking ibuprofen, or drinks 15 litres instead of 1.5, she is back in hospital within the week.



THE EVIDENCE

Discharge is where instructions get lost

48%

of patients knew what their medicines were for

Karliner et al., Medical Care 2012

50%

of limited-English patients knew their follow-up (vs 66%)

Karliner et al., Medical Care 2012

Grade 10

reading level of discharge notes; the AMA recommends grade 6

Mayo Clinic, Am J Surg 2016

−59%

post-discharge calls after rewriting notes more simply (22 → 9 per 100)

Mayo Clinic, ACS 2017

WHY AI ALONE IS NOT THE ANSWER

Rewriting is easy. Trusting the rewrite is the problem.

18%

of physician reviews of GPT-4 patient-friendly summaries raised a safety concern

Zaretsky et al., JAMA Netw Open 2024

24%

found omissions; 4% found hallucinations

Zaretsky et al., JAMA Netw Open 2024

"keep taking"

is what machine translation made of "hold the kidney medicine" in Chinese

Khoong et al., JAMA Intern Med 2019

HOW HOMEWARD WORKS

Every sentence earns its place

1 · Mask

Names, record numbers, birth dates, phones replaced before any model call

2 · Fact ledger

Every instruction tied to an exact quote; medicine list cross-checked by rules

3 · Plain draft

Grade 6, each sentence cites its facts

4 · Translate

…and back-translate to check meaning

5 · Check

Numbers, medicines, stop vs keep, contacts, coverage; safety model verdict

6 · Human

Only flagged and high-risk lines; named sign-off

7 · Teach-back

Quiz from approved sentences; wrong answers alert the nurse

Five schema-constrained model calls per packet, all on one AMD Instinct MI300X. Deterministic checks around the model, so its mistakes are caught instead of trusted.

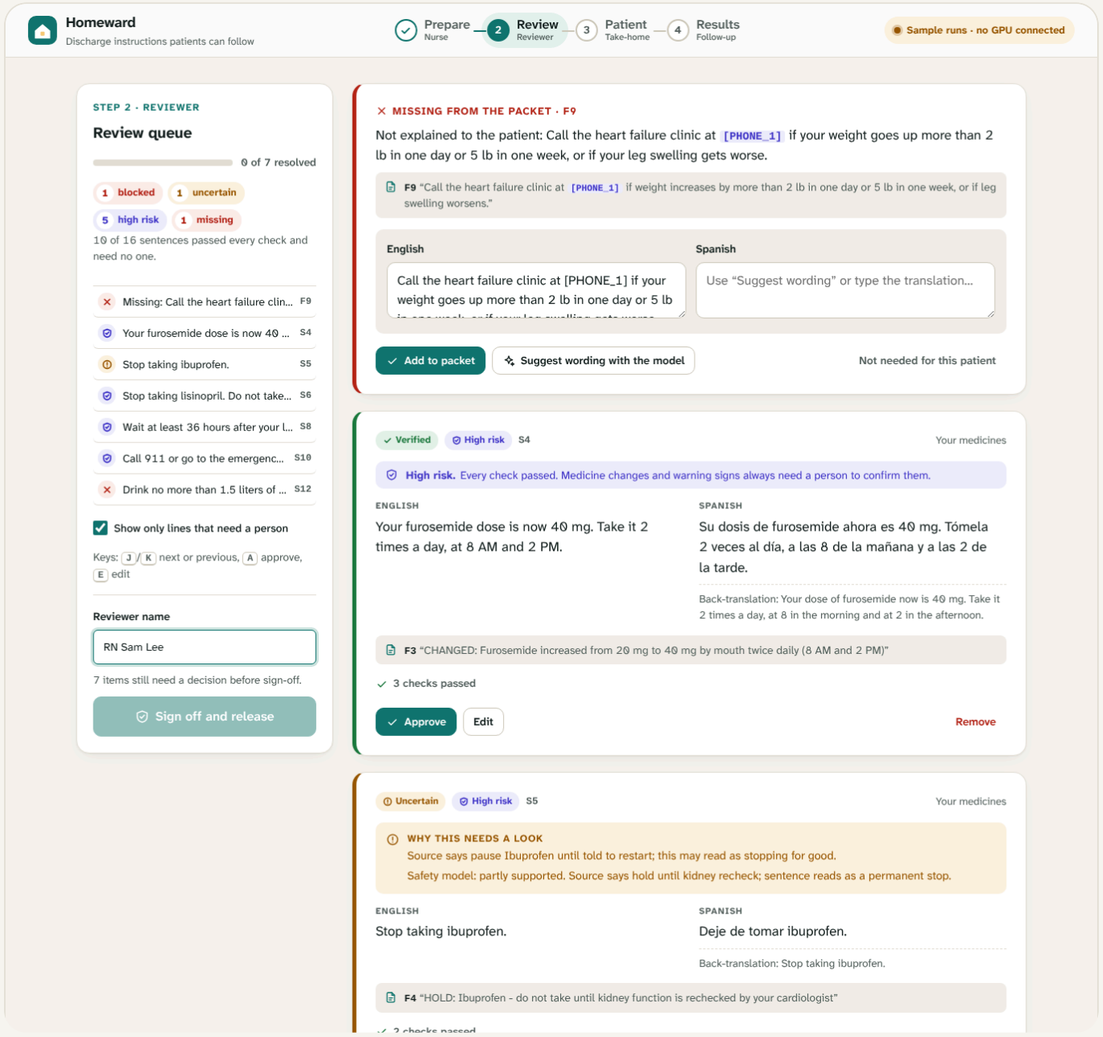
SAFETY, AS THE TRACK ASKS

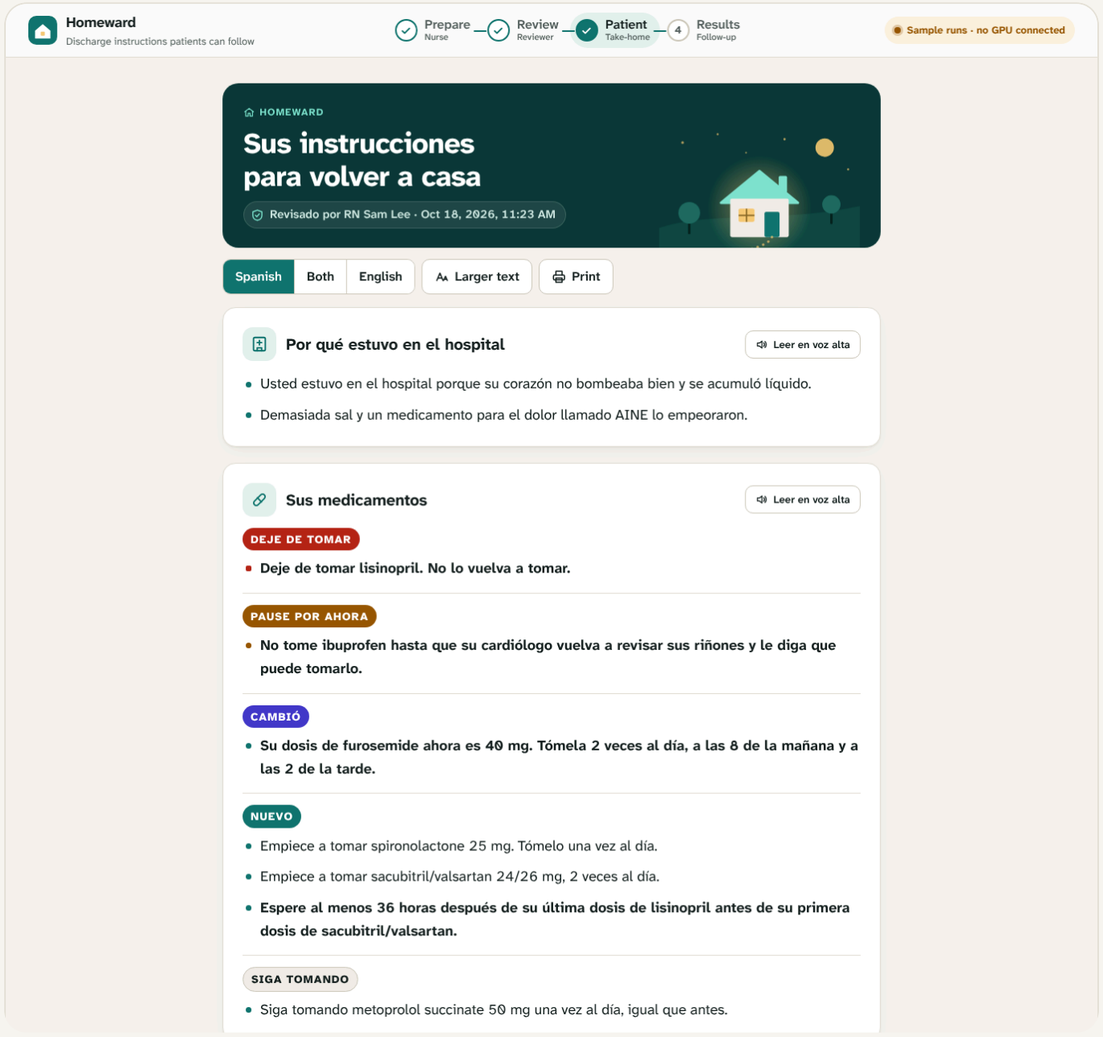
Six rules, built into the workflow

REQUIREMENT	HOW IT IS HANDLED
No diagnosis or advice	Restates the clinician only. No cited fact or an unknown medicine means blocked.
Wrong or unsupported output	Quote check, rule-based medicine cross-check, number / medicine / stop-vs-keep checks, safety model.
Sensitive information	Identifiers masked before any model call; self-hosted model; memory-only cases.
Uncertain results	green amber red on every sentence, with the reason. Unsupported language or too many failures go to an interpreter.
High-risk decisions	Blood thinners, insulin, opioids, any stop or dose change, warning signs: always signed off by a person.
Human review	Nothing reaches the patient without a named sign-off. Red lines cannot be approved as they are. Full audit trail.

THE PRODUCT

Review in minutes, not a full re-read





RESULTS

What it caught

8.7 → 4.1

reading grade, heart failure case

Flesch-Kincaid on the English draft

63%

of sentences verified with no human needed

the reviewer reads 6 of 16

3 / 3

planted problems stopped: 15 L vs 1.5 L, hold-as-stop, missing warning

plus an invented ibuprofen in the Italian case

100%

of 162 injected errors flagged by the deterministic checks alone

0 false alarms on 33 clean sentences; synthetic test

AMD

A 72B multilingual model on one GPU, inside the hospital

- vLLM + ROCm on an AMD Instinct MI300X (192 GB HBM3), AMD Developer Cloud
- 192 GB fits Qwen2.5-72B-Instruct in bf16 on a single GPU: no third-party API sees patient text
- vLLM batches a whole ward's discharges on one card
- Measured on MI300X: latency, tokens/s and GPU cost per packet (from scripts/benchmark.py)

VALUE AND NEXT STEPS

Fewer calls back, fewer readmissions, less nurse time

- Every discharge, every language, at GPU cost of cents per packet
- Reviewer time goes to the 4 in 10 sentences that need it
- Teach-back catches misunderstanding before the patient leaves, not after
- Next: FHIR read from the EHR, more languages, pilot on one ward with nurse time and call-back rates measured

Homeward never diagnoses. It makes sure what the clinician wrote reaches the patient, in their language, and that they understood it.